

## WARWICK, Australia

WMO: 945550

Lat: 28.2061S

Lon: 152.1003E

Elev: 476

StdP: 95.74

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 7             | -1.1       | 0.6 | -7.4                          | 2.1 | 13.2 | -4.9 | 2.6 | 14.1 | 8.6                   | 14.6 | 7.6 | 15.1 | 0.3                   | 60   | 0.406 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 1             | 11.6                   | 34.0            | 20.3 | 32.3 | 20.0 | 30.7 | 19.6 | 23.2                | 29.4 | 22.2 | 28.0 | 21.5 | 27.1 | 4.5                  | 320  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 21.6                            | 17.2 | 25.3 | 20.6 | 16.1 | 24.4 | 19.8 | 15.4 | 23.8 | 71.4          | 29.4 | 67.6 | 28.0 | 65.0 | 27.2 | 35.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 8.1           | 7.1  | 6.3 | -4.2                       | 37.3 | 1.4                | 1.8 | -5.1                                               | 38.6 | -5.9       | 39.6 | -6.7       | 40.6 | -7.7       | 41.9 |    |    |
| (5)               |      |     | -2.5                       | 26.4 | 1.8                | 4.0 | -3.8                                               | 29.3 | -4.8       | 31.6 | -5.8       | 33.8 | -7.1       | 36.7 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 17.5   | 23.4  | 22.8  | 21.3  | 17.9  | 14.0  | 11.6  | 10.7  | 11.7  | 15.2  | 18.2  | 20.8  | 22.4  |  |  |
|                                                                        | DBStd       | 5.19   | 2.28  | 2.38  | 1.93  | 2.19  | 2.56  | 2.64  | 2.46  | 2.64  | 2.77  | 2.77  | 2.85  | 2.66  |  |  |
|                                                                        | HDD10.0     | 51     | 0     | 0     | 0     | 0     | 2     | 14    | 22    | 12    | 1     | 0     | 0     | 0     |  |  |
|                                                                        | HDD18.3     | 962    | 0     | 1     | 1     | 32    | 135   | 202   | 238   | 207   | 99    | 37    | 9     | 2     |  |  |
|                                                                        | CDD10.0     | 2779   | 416   | 359   | 351   | 238   | 127   | 63    | 42    | 63    | 157   | 253   | 323   | 386   |  |  |
|                                                                        | CDD18.3     | 649    | 158   | 127   | 93    | 20    | 1     | 0     | 0     | 1     | 7     | 32    | 81    | 129   |  |  |
|                                                                        | CDH23.3     | 5681   | 1359  | 1014  | 674   | 171   | 14    | 1     | 1     | 25    | 156   | 395   | 768   | 1103  |  |  |
|                                                                        | CDH26.7     | 1939   | 527   | 369   | 180   | 20    | 1     | 0     | 0     | 6     | 33    | 110   | 277   | 417   |  |  |
| (14) Wind                                                              | WSAvg       | 2.7    | 3.2   | 3.0   | 2.7   | 2.4   | 2.1   | 2.3   | 2.2   | 2.4   | 2.7   | 2.8   | 3.2   | 3.2   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 728    | 85    | 84    | 64    | 39    | 53    | 28    | 44    | 32    | 34    | 70    | 80    | 111   |  |  |
|                                                                        | PrecMax     | 1105   | 194   | 266   | 200   | 308   | 249   | 100   | 144   | 91    | 105   | 196   | 183   | 284   |  |  |
|                                                                        | PrecMin     | 446    | 5     | 7     | 3     | 0     | 0     | 1     | 0     | 0     | 0     | 15    | 15    | 22    |  |  |
|                                                                        | PrecStd     | 156    | 48    | 56    | 54    | 54    | 49    | 22    | 38    | 22    | 26    | 43    | 46    | 61    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 35.6  | 35.9  | 33.3  | 28.8  | 25.1  | 21.8  | 21.9  | 26.3  | 29.9  | 32.2  | 34.5  | 35.6  |  |  |
|                                                                        |             | MCWB   | 21.8  | 21.3  | 20.1  | 17.6  | 16.1  | 14.7  | 13.1  | 14.2  | 15.8  | 17.1  | 18.9  | 19.5  |  |  |
|                                                                        | 2%          | DB     | 33.4  | 32.7  | 30.4  | 26.8  | 23.0  | 20.1  | 19.9  | 22.8  | 27.2  | 29.5  | 32.1  | 33.1  |  |  |
|                                                                        |             | MCWB   | 20.7  | 21.5  | 19.8  | 17.2  | 15.0  | 13.8  | 12.1  | 12.8  | 14.9  | 16.6  | 18.2  | 19.9  |  |  |
|                                                                        | 5%          | DB     | 31.5  | 30.7  | 28.6  | 25.3  | 21.6  | 18.7  | 18.5  | 20.8  | 25.0  | 27.4  | 29.8  | 30.9  |  |  |
|                                                                        |             | MCWB   | 20.6  | 20.7  | 19.4  | 17.0  | 14.3  | 12.9  | 11.7  | 12.0  | 14.3  | 16.4  | 18.0  | 19.6  |  |  |
|                                                                        | 10%         | DB     | 29.6  | 28.8  | 27.1  | 23.9  | 20.2  | 17.4  | 17.1  | 19.0  | 22.8  | 25.4  | 27.7  | 29.0  |  |  |
|                                                                        |             | MCWB   | 20.3  | 20.4  | 18.9  | 16.6  | 13.9  | 12.3  | 11.2  | 11.3  | 13.7  | 15.9  | 17.7  | 19.4  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 26.5  | 25.2  | 22.6  | 20.7  | 18.8  | 16.7  | 16.6  | 16.2  | 18.1  | 21.0  | 21.7  | 25.0  |  |  |
|                                                                        |             | MCDB   | 29.9  | 30.4  | 28.3  | 25.4  | 21.0  | 19.4  | 19.2  | 22.4  | 24.7  | 26.2  | 29.4  | 26.1  |  |  |
|                                                                        | 2%          | WB     | 23.1  | 23.1  | 21.5  | 19.2  | 17.0  | 15.2  | 14.7  | 14.6  | 16.9  | 19.1  | 20.6  | 22.1  |  |  |
|                                                                        |             | MCDB   | 29.4  | 29.3  | 26.5  | 23.7  | 20.9  | 18.1  | 17.4  | 19.6  | 23.1  | 25.2  | 27.5  | 28.4  |  |  |
|                                                                        | 5%          | WB     | 22.1  | 22.1  | 20.7  | 18.3  | 15.9  | 14.2  | 13.3  | 13.5  | 15.8  | 18.0  | 19.7  | 21.2  |  |  |
|                                                                        |             | MCDB   | 28.2  | 27.4  | 25.7  | 22.9  | 19.4  | 17.0  | 16.4  | 18.4  | 21.7  | 24.0  | 26.2  | 27.1  |  |  |
|                                                                        | 10%         | WB     | 21.3  | 21.3  | 19.9  | 17.4  | 15.0  | 13.3  | 12.2  | 12.4  | 14.9  | 17.0  | 18.9  | 20.5  |  |  |
|                                                                        |             | MCDB   | 26.9  | 26.5  | 24.6  | 22.0  | 18.6  | 16.1  | 15.6  | 17.5  | 20.9  | 22.7  | 24.7  | 26.0  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 11.6  | 11.2  | 11.5  | 12.1  | 12.9  | 11.9  | 13.5  | 15.3  | 15.0  | 13.6  | 12.5  | 11.8  |  |  |
|                                                                        |             | MCDBR  | 15.3  | 14.8  | 14.7  | 14.8  | 15.3  | 14.1  | 16.2  | 18.8  | 19.1  | 17.6  | 16.2  | 15.5  |  |  |
|                                                                        |             | MCWBR  | 4.8   | 4.8   | 5.2   | 6.2   | 7.7   | 7.8   | 8.5   | 9.3   | 8.1   | 6.7   | 5.4   | 5.0   |  |  |
|                                                                        | 5% WB       | MCDBR  | 12.0  | 12.0  | 11.2  | 11.8  | 11.1  | 10.4  | 11.2  | 14.1  | 14.0  | 13.5  | 12.6  | 11.8  |  |  |
|                                                                        |             | MCWBR  | 5.1   | 5.0   | 4.5   | 5.6   | 6.4   | 6.4   | 7.1   | 8.1   | 6.9   | 6.3   | 4.9   | 5.1   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.361 | 0.350 | 0.331 | 0.313 | 0.292 | 0.287 | 0.279 | 0.292 | 0.330 | 0.350 | 0.363 | 0.367 |  |  |
|                                                                        | taud        |        | 2.529 | 2.574 | 2.624 | 2.645 | 2.650 | 2.666 | 2.675 | 2.612 | 2.492 | 2.471 | 2.478 | 2.497 |  |  |
|                                                                        | Ebn at Noon |        | 981   | 974   | 964   | 935   | 913   | 897   | 920   | 945   | 949   | 963   | 972   | 977   |  |  |
|                                                                        | Edh at Noon |        | 112   | 104   | 94    | 83    | 75    | 70    | 72    | 84    | 105   | 114   | 117   | 116   |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 6.71  | 6.11  | 5.26  | 4.56  | 3.74  | 3.10  | 3.54  | 4.48  | 5.51  | 6.18  | 6.67  | 6.72  |  |  |
|                                                                        | RadStd      |        | 0.55  | 0.55  | 0.32  | 0.21  | 0.23  | 0.23  | 0.27  | 0.33  | 0.45  | 0.60  | 0.64  | 0.67  |  |  |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (47) | Regional (28 neighbors) | +0.41 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | -65     | +180    | +110    |

Nomenclature: See separate page